

Online Digital pH Sensor User Manual

Xi'an Desun Uniwill Electronic Technology Co., Ltd.

 Please read the operating instructions carefully before use



Preface

Thank you very much for choosing our company!

Please read this manual carefully before using this product, and save it for reference. Please observe the operating procedures and precautions in this manual.

To ensure that the after-sale protection provided by this instrument is effective, please do not use and maintain the instrument in any way other than specified in this manual.

As the failure to observe the precautions specified in this operation manual, any failures and losses caused are not covered by the manufacturer's warranty, and the manufacturer does not assume any related responsibility. Please keep all documents in a safe place. If in doubt, please contact our after-sales service department.

When receiving the instrument, please carefully open the package and check whether the instrument and accessories are damaged due to shipping. If damage is found, please contact Desun Uniwill after-sales service department and keep the packaging for return.

When the instrument fails, please do not repair it by yourself, please contact our after-sales service department.

Online Digital pH Sensor

I. Overview

Our digital pH sensor adopts industrial online electrode and built-in temperature sensor, which can automatically compensate for temperature, which is suitable for online long-term monitoring environment. The sensor adopts RS485 output and supports Modbus, which can realize networking and system integration without a controller.



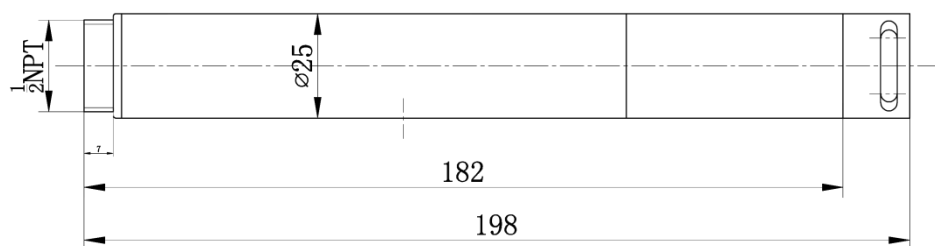
Features

- ✧ Composite electrode technology, long service life
- ✧ Stable, independent of transmission distance
- ✧ Built-in temperature sensor, automatic temperature compensation
- ✧ Network integration without controller
- ✧ RS485 output

Technical Parameter

Item	Parameters
Model	DS510
Range	0~14pH
Precision	0.01
Resolution	0.01
Deepest depth	Underwater 30m
Protection level	IP68
Maximum operating pressure	3bar
Storage temperature	0~ 60 °C
Temperature range	0~ 50°C
Sensor interface	Support RS-485, MODBUS protocol
Assembly	Input type
Power information	DC 5~12V, current <50mA
Size	Φ 25 * 198 mm
Probe cable length	5 meters (default), can be customized
Housing material	POM, SS316
Calibration	Three-point calibration

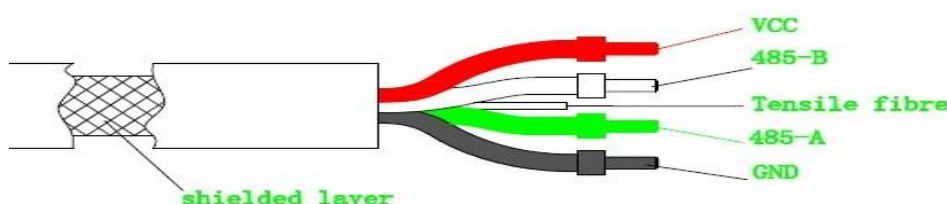
Dimensions



II. Cable information for Sensors

4 wire AWG-24 or AWG-26 shielding wire. OD=5mm

- Red line----power supply (VCC)
- White line---- 485 data -B (485-B)
- Yellow line---- 485 data -A (485-A)
- Black line----ground (GND)
- Bare line---- shielding layer



III. Installation

Please check the appearance of the cable for damage and disperse the cable before installation and operation.

- ① When using the input sensor, avoid the sensor hitting the wall or other facilities due to water flow, and fix the sensor if the water flow rate is too fast.
- ② There should be no obstruction in front of the measuring surface. pH sensor should be installed with the electrode protection cover to avoid damage to the glass bubble.
- ③ The small metal hole on the pH sensor should be submerged in water in order to measure accurately.
- ④ The water depth where the sensor is installed should not exceed 2m from the horizontal, and according to the fluctuation of water flow, the water depth at the installation location should not be less than 30cm. (Adjustable according to the body of water measured)
- ⑤ Should not be used to lift the sensor itself cable, the installation of cable protection devices to prevent the cable and the sensor between the taut tension is too large, resulting in cable breakage.
- ⑥ The sensor should not be installed with the measuring surface facing upwards, can be installed vertically downwards or horizontally.
- ⑦ If the product is used in the water diversion tank, please consult the company's sales staff to confirm the specific requirements for installation.

IV. Maintenance Methods and Common Problems

Maintenance Schedule and Method

Maintenance Schedule

It is recommended to periodically calibrate or clean the sensor surface according to the actual application scenario. The calibration sequence for the three-point calibration is that the first point calibration is at pH 4.0, the second point calibration is at pH 6.86, and the third point calibration is at pH 9.18 in a buffer solution. Refer to the MODBUS document for specific calibration methods.

Maintenance tasks,	Recommended maintenance frequency
Calibrate the sensor (if required by the competent authority)	According to the maintenance schedule required by the competent authority

Maintenance Method

Sensor Maintenance

- 1) **External surface of the sensor:** Wash the external surface of the sensor with tap water. If debris remains, wipe it with a moist soft cloth.
- 2) **Check the cable of the sensor:** The cable should not be tight during normal work, otherwise the internal wires of the cable will be easily broken and the sensor will not work properly.
- 3) **Sensor storage:** When the sensor is not in use, you should fasten the black plastic cap and check whether the sponge inside is wet. If it is not wet enough, please add 3mol / L potassium chloride solution and let the electrode be stored in the solution with potassium chloride.

Precautions

The probe contains sensitive electronic components. Ensure that the probe is not subjected to severe mechanical shock. Damage to the sensor caused by impact or human effort will not be covered by the warranty. Water ingress to the sensor due to replacement of the pH electrode is not covered by the warranty.

Common Problems

Error	Possible causes	Solutions
The operation interface cannot be connected or the measurement results are not displayed	Controller and cable connection error	Reconnect the controller and cables
	Cable failure	Please contact us
Measured values are too high, too low, or the values remain unstable	The metal holes of the sensors are not submerged in water	Submerge the sensor completely in water
	The sensor is not fixed	Fixing the pH sensor in place
	Water has entered the product	Please contact us
	The pH electrolyte is exhausted	Replace the pH electrochemical part
	Measurement shifted	Recalibration

V. Calibration of Sensors

Start pH Calibration	The pH electrode can support three-point calibration. The standard buffer solution is			
	4.00	6.86	9.18	(25℃)
The calibration sequence of the three-point calibration is that the first calibration is at pH 4, the second calibration is at pH 6.86, and the third calibration is in a buffer solution of pH 9.18. The calibration process is: Step 1: Immerse the electrode in the first calibration standard for at least 1 minute. Step 2: Begin Calibration: User enters pH value for this calibration standard 01 10 12 00 00 02 04 00 00 80 3F CRC 01 10 12 00 00 02 4A 4C Step 3: Repeat the above two steps to calibrate the second and third points Step 4: Read the calibration status to determine whether the calibration is successful.				
Criteria for successful calibration	The pH value is stable for more than 1 minute; The currently measured pH value matches one of the calibration standard solutions; Calibration parameters should be within a reasonable range.			

VI. Quality Assurance

Dear Customer:

First of all, thank you for choosing to buy our products!

In order to guarantee the quality of our products, to clarify the responsibility of both the buyer and the seller for the quality of our products, to ensure the quality of our products and to guarantee the safety of our products, we would like to make the following guarantee.

The pH sensors provided by our company are guaranteed for one year from the date of delivery.

Upon receipt of a product quality problem from a customer, we will confirm within 24 hours whether the product requires repair; do not send the product back without permission for repair.

In principle, the quality of the products provided by our company will be in accordance with the technical standards stipulated by the purchaser. If the purchaser does not specify the technical standards, we will follow the current national standards or implement the relevant factory standards, and ensure the stability and gradual improvement of product quality.

The supplier guarantees that the products supplied meet the statutory quality standards and is responsible for the quality of the products, and that the product packaging and registered trademarks comply with the relevant national regulations.

The purchaser is responsible for the quality of the products due to poor storage and maintenance of the products.

VII. Warranty limit

The warranty does not cover the following:

- Damage caused by natural disasters, riots, war and other force majeure factors.
- Damage caused by improper use, negligence, accidents or improper use by the user.
- Shipping costs incurred for the return of the product.
- Costs incurred for postage of product parts during the warranty period.
- Travel costs for out-of-warranty repairs.

The remedies such as repair, replacement, or refund of the payment as described above are special cases that do not violate this warranty. The remedies such as replacement or refund of the payment are for our products. Based on strict liability or other legal theories, our company is not responsible for any other damage caused by product defects or negligent operation, including the subsequent damage caused by causality with these situations.